

Double-click (or enter) to edit

```
!pip install transformers datasets torch accelerate
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.10/dist-packages (4.34.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (2.18.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.2.0)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (0.27.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (3.12.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (1.3.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (1.26.4)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (24.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (6.0.1)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (2023.12.31)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (0.22.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (0.12.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (0.5.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (4.66.1)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (15.0.2)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.1.4)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (2.32.2)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (3.5.0)
Requirement already satisfied: multiprocessing in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (2024.10.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (4.11.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (75.1.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (3.2.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (3.1.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (5.9.8)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (3.9.5)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (1.4.3)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (0.27.0)
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Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (3.3.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (2024.7.4)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (2.1.5)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (2024.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (2024.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (2.5.0)
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Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (25.0.1)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (1.4.1)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (6.0.5)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (0.2.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (1.17.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (4.3.0)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (1.0.5)
```

```
Requirement already satisfied: httplib2 in /usr/local/lib/python3.12/
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/c
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/di
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.1
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/pyt
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.1
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/p
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/li
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/
```

Double-click (or enter) to edit

```
text = """
Artificial Intelligence is transforming industries worldwide.
Machine learning enables computers to learn from data.
Deep learning powers modern applications such as image recognition.
AI is used in healthcare, finance, education, and transportation.
Generative AI can create text, images, music, and videos.
"""

with open("data.txt", "w") as f:
    f.write(text)

    print("Dataset created successfully")
```

Dataset created successfully

```
from datasets import Dataset

with open("data.txt", "r") as f:
    text = f.read()

    dataset = Dataset.from_dict({"text": [text]})
    dataset
```

```
from transformers import GPT2Tokenizer, GPT2LMHeadModel

tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
model = GPT2LMHeadModel.from_pretrained("gpt2")

print("GPT-2 loaded successfully")
```

```
Loading weights: 100% 148/148 [00:00<00:00, 237.75it/
s, Materializing param=transformer.wte.weight]

GPT2LMHeadModel LOAD REPORT from: gpt2
Key | Status | |
-----+-----+---
h.{0...11}.attn.bias | UNEXPECTED | |
```

Notes:

- UNEXPECTED :can be ignored when loading from different task/architec
- GPT-2 loaded successfully

```
tokenizer.pad_token = tokenizer.eos_token
```

```
def tokenize_function(examples):
    tokenized_inputs = tokenizer(
        examples["text"],
        truncation=True,
        padding="max_length",
        max_length=128
    )
    tokenized_inputs["labels"] = tokenized_inputs["input_ids"].copy()
    return tokenized_inputs

tokenized_dataset = dataset.map(tokenize_function, batched=True)

print(tokenized_dataset)
```

```
Map: 100% 1/1 [00:00<00:00, 12.47 examples/s]
```

```
Dataset({
  features: ['text', 'input_ids', 'attention_mask', 'labels'],
  num_rows: 1
```

```
tokenizer.pad_token = tokenizer.eos_token
model.resize_token_embeddings(len(tokenizer))
```

```
Embedding(50257, 768)
```

```
from transformers import Trainer, TrainingArguments, DataCollatorForLanguageModeling

data_collator = DataCollatorForLanguageModeling(tokenizer=tokenizer, mlm_probability=0.15)

training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=5,
    per_device_train_batch_size=1,
    save_steps=100,
    save_total_limit=1
)
```

```
trainer = Trainer(
```

```

    model=model,
    args=training_args,
    train_dataset=tokenized_dataset,
    data_collator=data_collator
)

```

```

# Temporarily inspect a batch from the data collator
# from torch.utils.data import DataLoader
# dataloader = DataLoader(tokenized_dataset, batch_size=1, collate_fn=d
# batch = next(iter(dataloader))
# print(batch.keys())
# print({k: v.shape for k, v in batch.items()})

trainer.train()

```

```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:
    super().__init__(loader)

```

 [5/5 00:42, Epoch 5/5]

Step Training Loss

Writing model shards: 100%

1/1 [00:01<00:00, 1.94s/

it]

```

TrainOutput(global_step=5, training_loss=2.6250194549560546,
metrics={'train_runtime': 63.8966, 'train_samples_per_second': 0.078,
'train_steps_per_second': 0.078, 'total_flos': 326615040000.0,
'train_loss': 2.6250194549560546, 'epoch': 5.0})

```

prompt = "Artificial Intelligence"

inputs = tokenizer(prompt, return_tensors="pt")

outputs = model.generate(**inputs, max_length=150, do_sample=True,
temperature=0.8, top_k=50, top_p=0.95)

```

generated_text = tokenizer.decode(outputs[0], skip_special_tokens=True)

print(generated_text)

```

```

model.save_pretrained("my_gpt2")
tokenizer.save_pretrained("my_gpt2")

```

Writing model shards: 100%

1/1 [00:16<00:00, 16.6/s/

it]

```
print(tokenized_dataset.features)
print(tokenized_dataset[0])
```

```
{'text': Value('string'), 'input_ids': List(Value('int32')), 'attention_mask': List(Value('int32'))}
{'text': '\nArtificial Intelligence is transforming industries worldwide'}
```